

Autonomous Systems Infrastructure Architecture Proposal

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Section 1: Executive Summary

This document details the architectural design and hardware specifications for the cloud-native, border-driven multi-agent orchestration infrastructure. Our plan ensures maximum resource density and high-contrast diagnostic capabilities.

Section 2: Hardware and API Infrastructure Thresholds

The physical infrastructure must utilize redundant 8x GPU nodes equipped with hardware-isolated memory contexts. Local networking links must support throughput speeds exceeding 400 Gbps to avoid processing bottlenecks during multi-agent workflows.

API communication channels must run over verified, secure gRPC tunnels with automatic rate-limiting and connection-pooling configurations.

Section 3: Security Compliance Parameters

All systems are subject to annual SOC 2 Type II audits, physical site reviews, and external penetration tests. Security telemetry must be streamed in real-time to the central Onyx Security Operations Center.

Section 4: Multi-Vendor Interaction Models

To prevent single-point failures, the infrastructure supports federated API dispatching. Ingested compliance items are processed across multiple independent model endpoints (Groq, Anthropic, and local Llama instances) using an automated model routing layer.